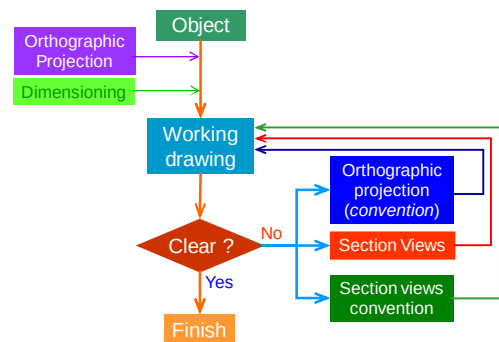


TOPICS

- Introduction
- Terminology & common practices
- Kind of sections
- Dimensioning

Introduction

GRAPHICS COMMUNICATION WITH ENGINEERING DRAWING

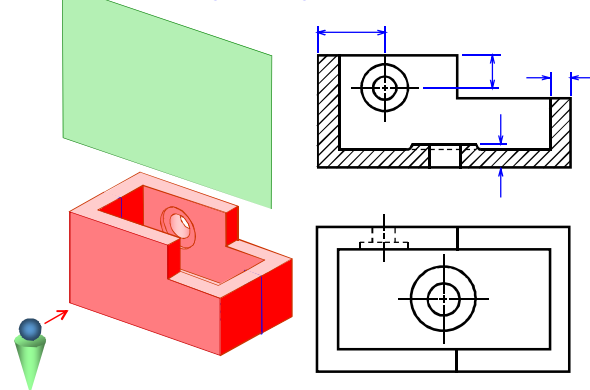


PURPOSES OF SECTION VIEWS

- Clarify the views by
 - ❖ reducing or eliminating the hidden lines.
 - ❖ revealing the cross sectional's shape.
- Facilitate the dimensioning.

Let See the

EXAMPLE : Advantage of using a section view.

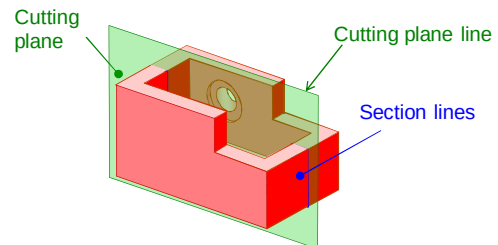


Terminology and common practices



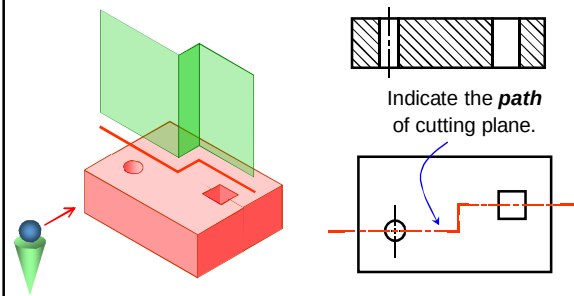
CUTTING PLANE

Cutting plane is a plane that *imaginarily cuts* the object to reveal the internal features.



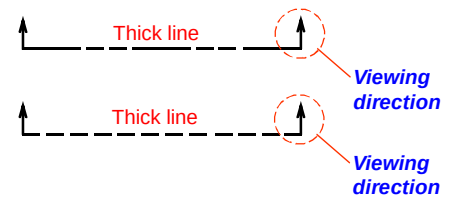
CUTTING PLANE LINE

Cutting plane line is an *edge view* of the cutting plane.

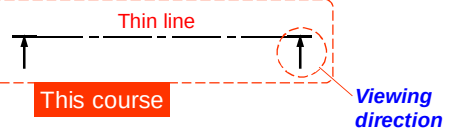


CUTTING PLANE LIFESTYLES

ANSI standard

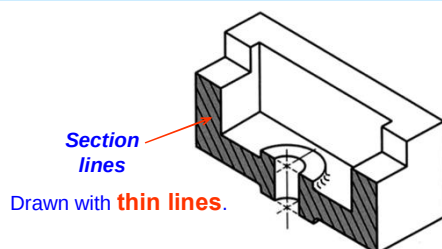


TS & ISO standard



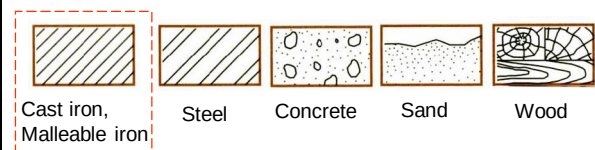
SECTION LINING

Section lines or *cross-hatch lines* are used to indicate the surfaces that are cut by the cutting plane.



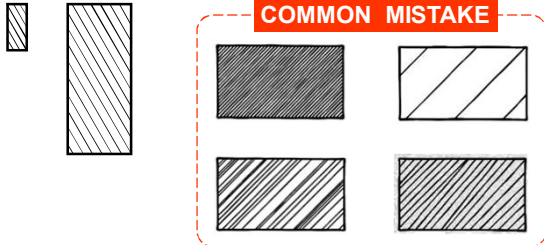
SECTION LINES SYMBOLS

- The section lines are different for each of material's type.
- For practical purpose, the cast iron symbol is used most often for any materials.



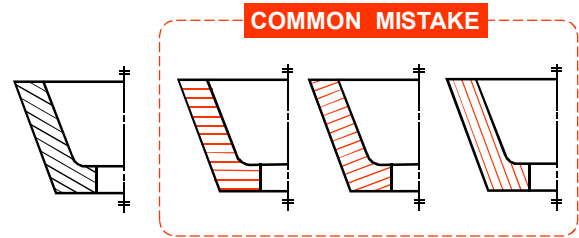
SECTION LINING PRACTICE

- The spaces between lines may vary from 1.5 mm for small sections to 3 mm for large sections.



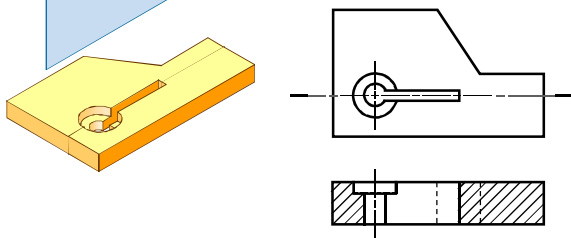
SECTION LINING PRACTICE

- It **should not** be drawn *parallel* or *perpendicular* to contour of the view.



TREATMENT OF HIDDEN LINES

- Hidden lines are **normally omitted** from section views.



Kinds of Sections

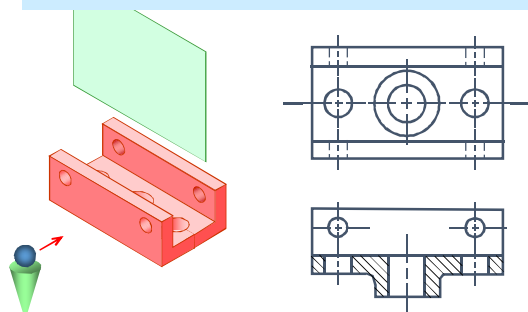


KIND OF SECTIONS

1. Full section
2. Offset section
3. Half section
4. Broken-out section
5. Revolved section (aligned section)
6. Removed section (detailed section)

FULL SECTION VIEW

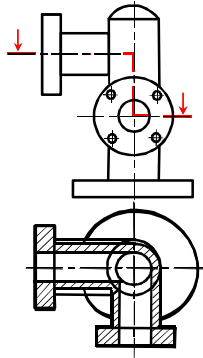
- The view is made by passing the **straight** cutting plane **completely through** the part.



OFFSET SECTION VIEW

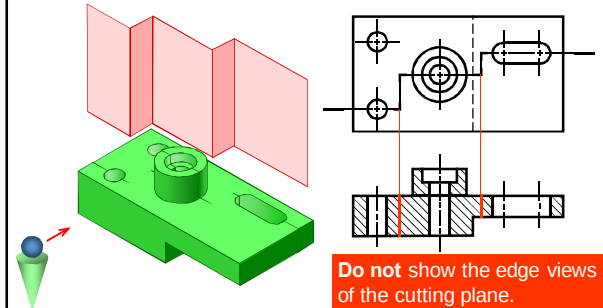
The cutting plane is *off-set* to include features that are not in a straight line.

It is possible for the cutting plane to *change directions*, to *minimise on the number* of sectional views required to capture the necessary details.



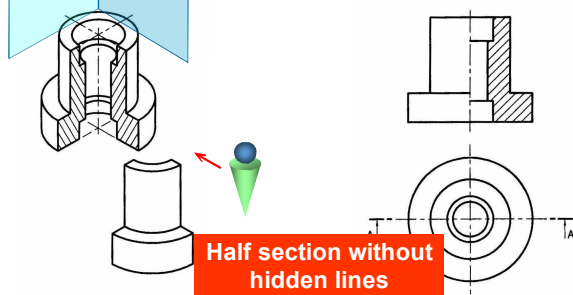
OFFSET SECTION VIEW

The view is made by passing the *bended* cutting plane *completely through* the part.



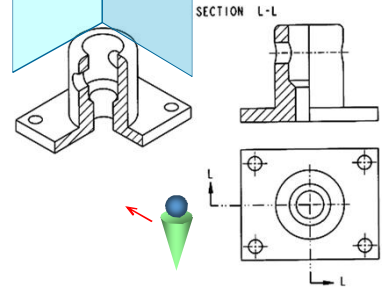
HALF SECTION VIEW

A half-section is a view of an object showing *one-half* of the view in section. Symmetrical parts can be shown in half sections.



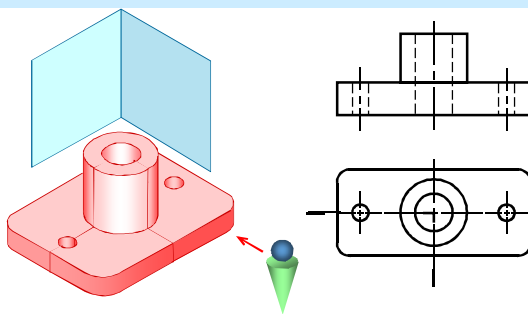
HALF SECTION VIEW

Half sections are commonly used to show both the *internal* and *outside* view of symmetrical objects.



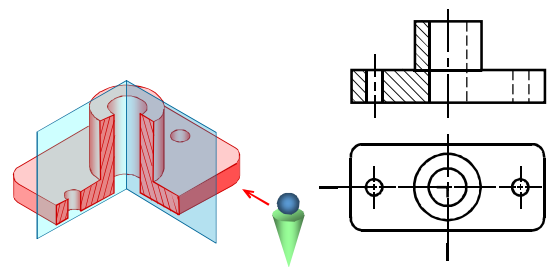
HALF SECTION VIEW

The view is made by passing the cutting plane *halfway* through an object and remove a *quarter* of it.



HALF SECTION VIEW

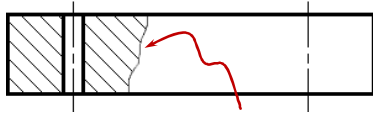
- A *center line* is used to separate the sectioned half from the unsectioned half of the view.
- *Hidden line* is omitted in unsection half of the view.



BROKEN-OUT SECTION VIEW

It is common practice to section a part of an object when only small areas need to be sectioned to indicate the important details

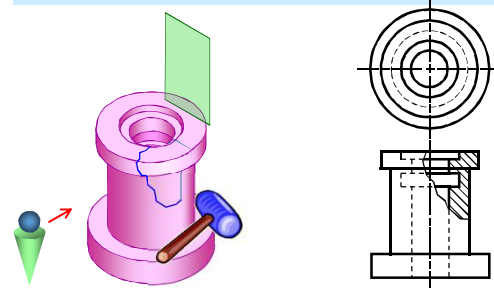
The example shows a **broken-out** sectional view to indicate a through-hole in a plate.



The lines indicating the end of the section is a thin continuous line

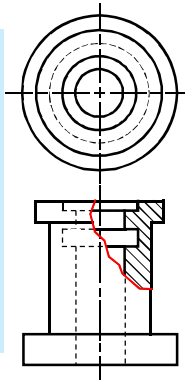
BROKEN-OUT SECTION VIEW

The view is made by passing the cutting plane normal to the viewing direction and removing the portion of an object in front of it.

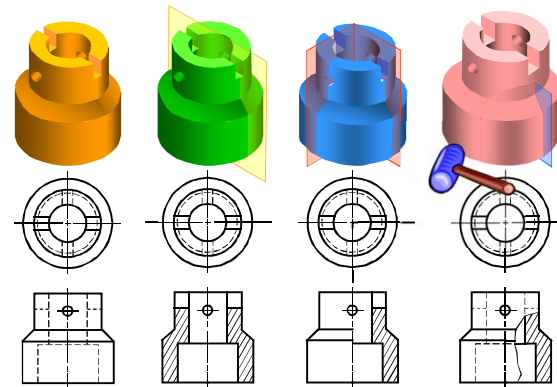


BROKEN-OUT SECTION VIEW

- A **break line** is used to separate the sectioned portion from the unsectioned portion of the view.
- Break line is a thin continuous line (0.25) and is drawn freehand.
- There is **no** cutting plane line.



EXAMPLE : Comparison among several section techniques

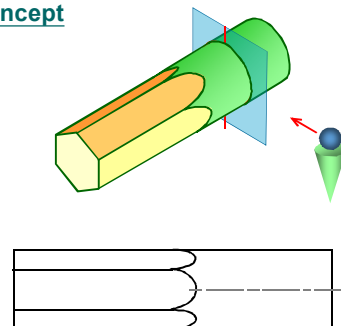


REVOLVED SECTION VIEW

- Revolved sections *show cross-sectional features* of a part.
- No need for additional orthographic views.
- This section is especially helpful when a cross-section varies.

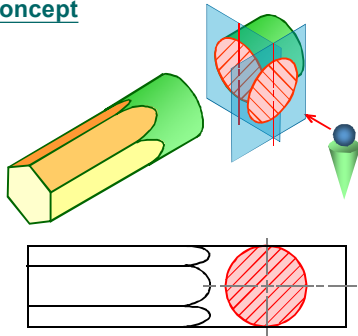
REVOLVED SECTION VIEW

Basic concept



REVOLVED SECTION VIEW

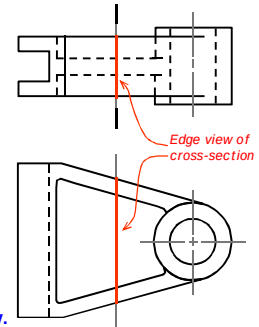
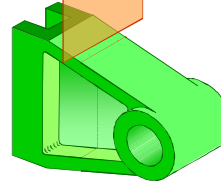
Basic concept



REVOLVED SECTION VIEW

Steps in construction

Given



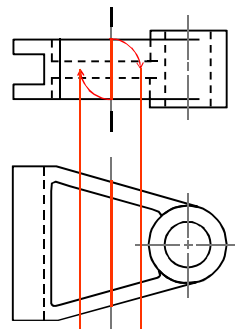
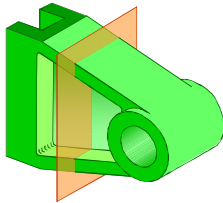
Step 1

- Assign position of cutting plane.
- Draw axis of rotation in front view.

REVOLVED SECTION VIEW

Steps in construction

Given



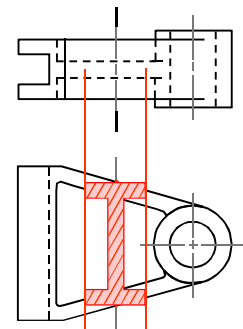
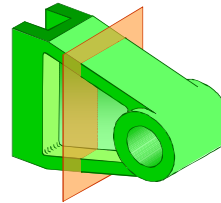
Step 2

- Transfer the depth dimension to the front view.

REVOLVED SECTION VIEW

Steps in construction

Given



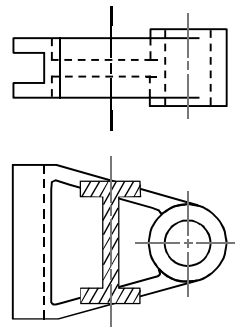
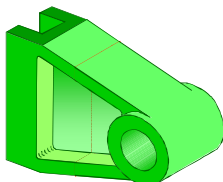
Step 3

- Draw the revolved section.
- Add section lines.

REVOLVED SECTION VIEW

Steps in construction

Given

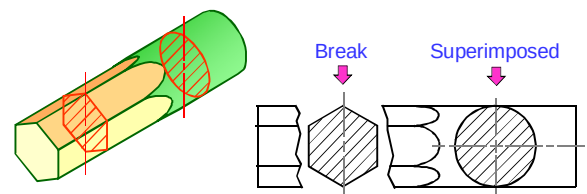


FINAL PICTURE

REVOLVED SECTION VIEW

Placement of revolved section

- Superimposed to orthographic view.
- Break from orthographic view.

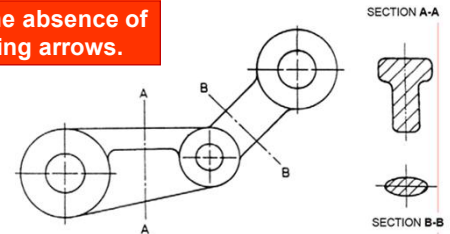


REMOVED SECTION VIEW

- Removed section is revolved section.
- Section view is shown outside the view.
- Used where space does not enough for revolved section
- Can be located elsewhere on a drawing with *properly labeled*
- It may be appropriate to use *removed sections*, for webs, beams or arms,

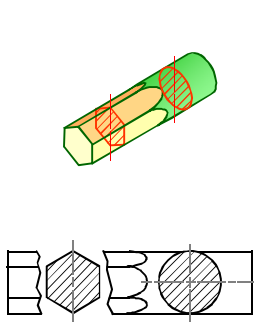
REMOVED SECTION VIEW

Note the absence of viewing arrows.

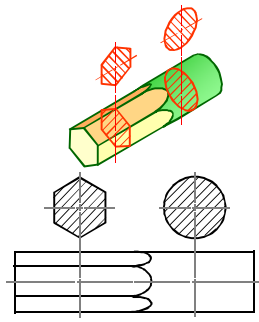


REMOVED SECTION VIEW

Revolved section

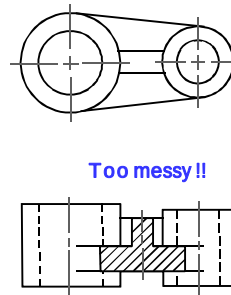


Removed section

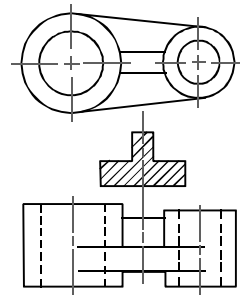


REMOVED SECTION VIEW

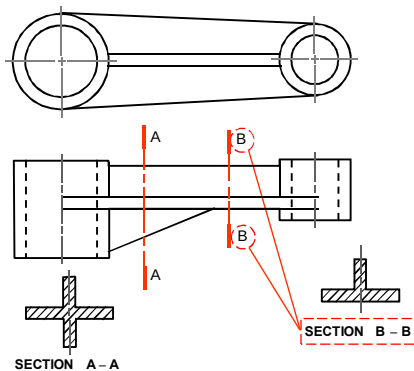
Poor



Preferred



REMOVED SECTION VIEW

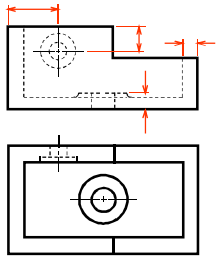


*Dimensioning
in Section View*

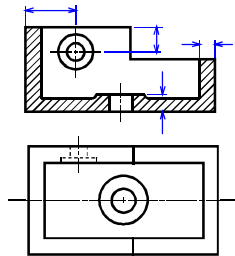


- In most cases, dimensioning of the section views follows the typical rules of dimensioning.

POOR

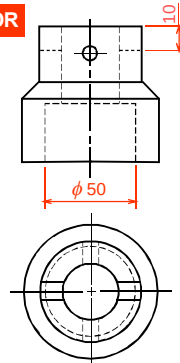


GOOD

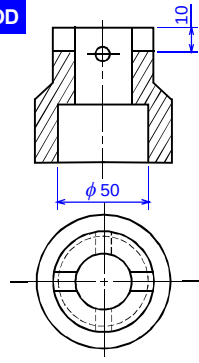


DIMENSIONING

POOR



GOOD



DIMENSIONING

- For a half-section view, use dimension line with only one arrowhead that points to the position inside the sectioned portion.

